

Thermal-Aware Test Scheduling with Floor planning for Three-Dimensional Stacked Integrated Circuit

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Abstract—Testing a three-dimensional stacked integrated circuit (3D-SIC) remains a challenging problem, as generating an optimized test schedule to minimize test time is complicated due to the numerous variables involved. Accessing upper dies is only feasible through the bottom die, necessitating the extension of Test Access Mechanisms (TAMs) via Through-Silicon Vias (TSVs). Limited primary I/O pins, TSVs, and TAM width require efficient resource allocation. Thermal management is crucial due to high core power consumption and uneven distribution, which pose the risk of overheating. Advanced concurrent test scheduling is essential to effectively allocate resources and maintain power and temperature limits. This research proposes thermal-aware test scheduling optimization combined with floor planning for 3D-SICs, aiming to minimize test schedule time while addressing resource and power constraints. Experimental results using several ITC'02 benchmark circuits demonstrate an average estimated improvement of 0.2% in test schedule time when utilizing test scheduling with floor planning compared to test scheduling without floor planning.

Keywords—3D-SIC, test scheduling, test time, floor planning, thermal-aware

I. INTRODUCTION

Integrating entire electronic systems onto a single chip is the prevailing trend. Despite advancements in integrated circuit (IC) technology enabling smaller feature sizes, conventional System-on-Chips (2D-ICs) are growing wider due to increased demand for semiconductor devices with enhanced features. However, this growth poses challenges to circuit performance due to lengthy horizontal interconnects between cores. To address this issue, three-dimensional stacked integrated circuit (3D-SIC) technology has emerged. It involves stacking circuit components into multiple layers with vertical interconnections (TSVs), thereby reducing horizontal interconnect lengths and offering improved functionality, bandwidth, and a smaller footprint [1-4].

Despite the advantages of 3D-SICs, testing is challenging due to vertically stacked dies. Test data transmission between the test pins and the cores on different dies, relies on test access mechanisms (TAMs) and TSVs [1]. However, due to the limited availability of test resources such as test pins,

TAMs, and TSVs, a test schedule with efficient allocation and utilization of these resources are essential, but finding an optimal solution is inherently complex [2-4]. Consequently, researchers use heuristics and approximation algorithms to create near-optimal test schedules, aiming to minimize test schedule time effectively [5-7].

Testing concurrently multiple cores in a 3D-SIC can reduce test schedule time but increase power consumption, risking system-level power limits and core damage [2-4]. Concurrent testing of adjacent cores with higher power density (hot modules) can create hotspots, compromising chip reliability [8]. Therefore, researchers frequently employ power and thermal-aware test scheduling strategies to manage test concurrency, ensuring that the overall power consumption and temperature of the 3D-SIC remain within system limits [8-10]. Furthermore, the placement of cores within the stack and their test duration impact heat dissipation capabilities. Cores further from the heat sink dissipate heat slower, regardless of their power consumption. Longer test times generate more heat, exacerbating potential hotspots. Therefore, test scheduling must consider core heat dissipation, ensuring adequate space around cores for heat dissipation to prevent hotspot formation and irreversible damage to the 3D-SIC [8-9].

Heat from hot modules transfers to cooler adjacent cores, so placing them next to non-hot modules can prevent hotspot formation and allow for more concurrent testing [11]. However, when positioned far from the heat sink, hot modules may dissipate heat less effectively, raising the 3D-SIC temperature. Ideally, hot modules should be closer to the heat sink, often at the top of the stack, but this increases costs due to additional TSVs. Therefore, relying solely on resource, power, and thermal-aware test scheduling is inadequate; thermal-aware floor planning that considers TAM placement is crucial for optimizing test schedule quality, an aspect often overlooked in previous research.

II. METHODOLOGY

A. Problem Formulation

Given a 3D-SIC to be designed with (i) the maximum allowable number of stacks, (ii) the maximum TAM width

limit, W_{max} , (iii) the maximum TSV limit, TSV_{max} , (iv) the number of cores, C_i , (v) the number of I/O terminals per C_i , (vi) the number of internal scan chains per C_i , (vii) the length of each internal scan chain, (viii) the number of test patterns associated with each C_i , (ix) maximum power limit, P_{max} , (x) maximum thermal limit, T_{max} . A co-optimization of thermal-aware test scheduling and floor planning techniques was performed to obtain a test schedule for 3D-SIC with minimal test time under P_{max} , T_{max} , and resource constraints.

The study's methodology, depicted in Fig. 1, illustrates the co-optimization process for thermal-aware test scheduling with floor planning, taking into account resource and power constraints. The main objective is to develop a test scheduling technique for 3D-SICs aimed at minimizing overall test time..

B. Pareto Test Cubes

To support modular testing and optimize the utilization of TAM widths and core I/O pins, several wrapper designs were created for each core using the Design Wrapper heuristic based on the Best Fit Decreasing (BFD) algorithm, as proposed in [12]. Test times for each wrapper design were calculated, resulting in a range of test times for each core across various TAM widths. Pareto-optimal points were identified based on these calculations. Each core under a specific Pareto optimal wrapper design was represented as a 3-dimensional cube, with dimensions corresponding to TAM width, power consumption, and test time.

C. Ant Colony Optimization

The Ant Colony Optimization (ACO) algorithm was used to select pareto cubes for 3D-SIC floor planning and test

scheduling. In ACO, each core was denoted as C_i ($1 \leq i \leq N$), where N represented the total number of cores in the 3D-SIC. Each core C_i was associated with a collection of pareto-optimal cubes (R_i). For each cube (r_i) in the set R_i , the dimensions are represented: the height as the allocated TAM width for the j th wrapper configuration, w_{ij} ; the width as the corresponding test time, $T(w_{ij})$; and the third dimension as the corresponding peak power, $P(w_{ij})$.

ACO leveraged both pheromone trail values and heuristic favourability to find solutions. In this study, the heuristic favourability, denoted as W_{prefer} , represented the preferred TAM width for C_i . Initially, the pareto cube from the set R_i with the highest TAM width, which represented the lowest test time, was selected as W_{prefer} for each C_i . Subsequently, the W_{prefer} value for each C_i was updated to its corresponding pareto cube from the set of cubes of the global best ant (R_{global}), which resulted in the lowest test schedule time, to prevent the search from becoming too focused on local optima. The pheromone trail value, denoted as $\tau(i, j)$, represented the attractiveness of choosing a pareto cube with TAM width j for C_i . The probability of selecting TAM width j for C_i was calculated based on (1), where α was the relative importance of $\tau(i, j)$, and β was the relative importance of TAM width, w_{ij} [13].

$$prob(i, j) = \frac{[\tau(i, j)]^\alpha \cdot \beta}{\sum_{j=1}^{W_{max}} ([\tau(i, j)]^\alpha \cdot \beta)} \quad (1)$$

In the ACO algorithm, a set number of ants were randomly placed within test cubes during initialization. Each ant selected test cubes for cores iteratively. After $N-1$ iterations, the ants had chosen a test cube for each core. These cubes were used for thermal-aware 3D-SIC floor planning, generating multiple floor plans. Each floorplan underwent thermal-aware test scheduling with power and TAM constraints, producing test schedule time. The set of cubes resulting in the shortest test schedule time was denoted as R_{global} , while R_{local} signified the optimal solution discovered during the current iteration of the ACO, primarily used for updating the pheromone trail. This process continued until reaching the maximum cycle count or stagnation, where all ants selected the same cubes with identical test times.

D. Thermal-aware 3D Floor planning

The set of cubes chosen by each ant undergoes a thermal-aware floor planning process using a greedy algorithm. The purpose of generating a floor plan for each ant is to explore different configurations and find the most optimal layout considering both thermal factors and TAMs. The inputs include the number of 3D-SIC layers and the maximum layer area, as well as information for each core such as its area (a_i), power density (p_{ij}), and TAM width (w_{ij}). Subsequently, the hot modules of the 3D-SIC are identified by calculating the average power density (p_{avg}) of all cores in the set. A core with a power density value greater than the average is designated as the hot module within the set. To mitigate hotspots and manage peak temperatures, special attention is given to the placement of the hot module, strategically arranging its position [11].

In this research, the optimal placement of the hot module involves several considerations. First, it is important to avoid

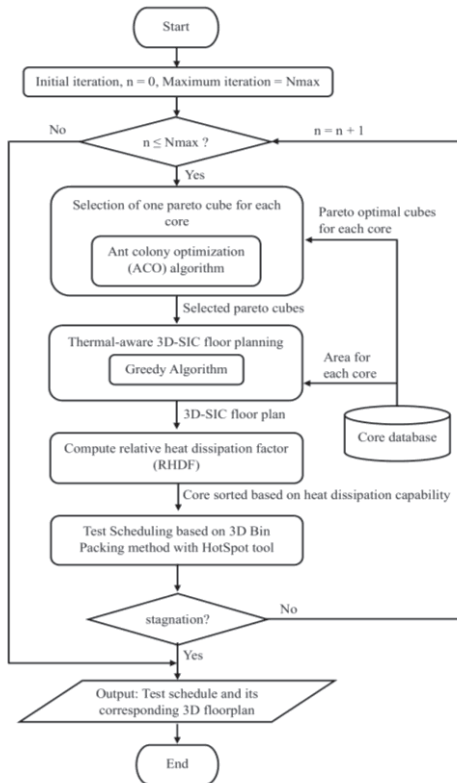


Fig. 1. Co-optimization procedure for thermal-aware test scheduling and floor planning in 3D-SICs

placing hot modules in close proximity to each other to prevent localized temperature build-up and potential thermal issues within the chip. Secondly, positioning the hot module closer to the heat sink is beneficial for efficient heat dissipation and helps maintain temperatures within acceptable limits. Lastly, placing the hot module at the edge of the layer can enhance heat dissipation by increasing exposure to ambient air. Adhering to these guidelines can improve the overall thermal performance and reliability of the 3D-SIC

Moreover, this research considers a scenario where the heat sink is placed on the top layer of the 3D-SIC. Some hot modules may necessitate a larger TAM width. Positioning these hot modules closer to the heat sink would increase the number of required TSVs for test access. Therefore, it is crucial to carefully balance the trade-off between power density and TAM width to achieve advantages in two key areas: reducing the overall temperature of the system and optimizing TSV utilization. A trade-off function (TFi) that describes this trade-off between TAM width (w_{ij}) and power density (p_{ij}) is then calculated for each hot module using (2), where γ and δ represent the relative importance of w_{ij} and p_{ij} , respectively.

$$TF_i = (\gamma * w_{ij}) + (\delta * p_{ij}) \quad (2)$$

E. Relative Heat Dissipation Factor

For each ant, parameters such as the number of layers in the 3D-SIC, the maximum surface area of each layer, the thickness of the 3D-SIC, thermal interface material (TIM) thickness, the thickness of each die layer, the placement of each core within the 3D-SIC, the area of each core, and the power trace value of each core are derived from the preceding subsections. These parameters serve as inputs for computing the Relative Heat Dissipation Factor (RHDF) for each C_i in the 3D-SIC. The RHDF values for each core are computed for each ant (each floorplan) based on (3) [8].

$$RHDF = HHDF * VHDF * \text{Power trace} \quad (3)$$

The Horizontal Heat Dissipation Function (HHDF) for each C_i in a layer is determined by measuring its distance from the four edges of the layer in the horizontal plane. The HHDF is computed as the sum of these distances, reflecting the core's heat dissipation capability in the horizontal direction. Conversely, the Vertical Heat Dissipation Function (VHDF) considers the layer's position within the stack. It quantifies heat dissipation by calculating the distance of each layer from the heat sink. The Relative Heat Dissipation Factor (RHDF) for each C_i is then derived from the product of HHDF and VHDF, providing an index that evaluates the core's heat dissipation capability based on its vertical and horizontal positioning within the stack. Additionally, the RHDF accounts for the core's power trace value and the duration of testing, represented by the number of clock cycles applied during testing. This comprehensive assessment ensures an accurate evaluation of each core's relative heat dissipation capability [8].

F. Thermal-aware Test Scheduling

After obtaining a set of cubes from the ACO algorithm and generating the 3D-SIC floorplan, the cores (cubes) for each ant are sorted in a non-increasing order based on their RHDF values from the previous subsection. This sorting step

prioritizes cores with lower heat dissipation capability, which may generate higher heat during testing mode, in the scheduling process. Subsequently, the 3D bin packing algorithm takes a set of cubes and the corresponding RHDF list for an ant, along W_{max} , P_{max} , and T_{max} as inputs. The primary objective of the algorithm is to optimize the scheduling of these cubes while ensuring that, throughout the generated schedule, the total TAM width usage does not exceed W_{max} , the total power consumption remains below P_{max} , and the maximum temperature remains below T_{max} .

The scheduling process begins by selecting the first cube from the RHDF list and placing it in a 3D bin. Subsequently, the next cube is chosen from the RHDF list, and the adjacency exclusion principle is applied. This principle ensures that the next cube can only be scheduled if it is not adjacent in the 3D-SIC floorplan to the previously scheduled cube within the same test session. This approach allocates sufficient space around the tested core to effectively dissipate generated heat, preventing heat accumulation and hotspot formation. If the next cube is adjacent to the previously scheduled cube within the same test session, it is placed in a temporary list to be scheduled in the subsequent test session.

Before scheduling the next selected cube based on the adjacency exclusion principle, the algorithm checks the availability of unallocated TAM width compared to the TAM requirement of the cube to determine if test concurrency is feasible. Next, it verifies the total power consumption to ensure that P_{max} is adhered to throughout the test session if the cube is scheduled. Additionally, the algorithm checks the maximum temperature of the test session using the HotSpot 6.0 tool [14] to ensure that T_{max} is adhered to throughout the test session if the cube is scheduled. If the cube satisfies W_{max} , P_{max} , and T_{max} , it is scheduled within the 3D bin. However, if the cube fails to meet any of these limits, it is placed in a temporary list for scheduling in the subsequent test session.

III. RESULT AND DISCUSSION

To implement and evaluate the proposed test scheduling and floor planning techniques for 3D-SICs, widely recognized ITC'02 SoC benchmark circuits were utilized [15]. These benchmarks were established as standardized representative designs commonly employed in integrated circuit testing research. This study utilized core information

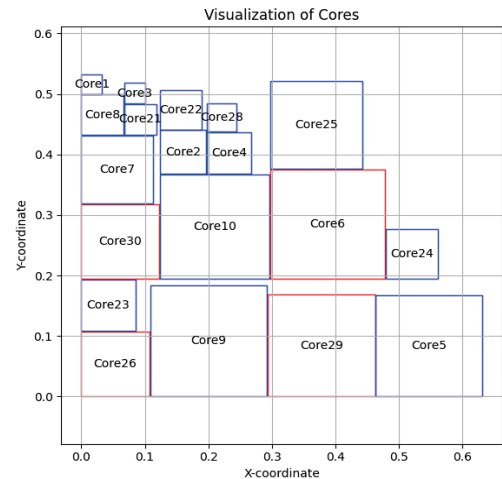


Fig. 2. A stack of 3D-SIC with cores from the p93791 SoC.

from four benchmark SOCs: d695 (an academic benchmark), and the industrial SOCs p22810, p34392 and p93791. The 3D-SIC and thermal parameters used in this research were adopted from [8].

The thermal-aware floor planning based on a greedy algorithm able to place hot modules such that hot modules are not adjacent to each other but are next to non-hot modules, as shown in Fig. 2, where cores with red borders are hot modules, and cores with blue borders are non-hot modules. Fig. 2 shows a stack of 3D-SIC with cores from the p93791 SoC. However, the thermal-aware floor planning process, which uses a greedy algorithm, cannot guarantee that all hot modules are placed at the edge of the layer. This limitation may affect the optimal heat dissipation capabilities of the system. The reliance on a greedy algorithm means it may not always find the globally optimal solution, potentially resulting in less efficient thermal management. Despite these challenges, the optimal placement strategy employed in this research still facilitates efficient heat dissipation during testing. It helps maintain the temperature within acceptable limits, as heat was effectively dissipated to the adjacent cores and the heat sink. Additionally, the strategic arrangement of hot modules, considering factors like proximity to the heat sink and avoidance of proximity to each other, contributes to improved thermal performance.

Since heat dissipation of hot modules were improved by thermal-aware floor planning, more cores was able to be tested concurrently, as the temperature of 3D-SIC able to remain under the system-level temperature limit. Hence, the overall test schedule time of a 3D-SIC was reduced. Table 1 shows a comparison of test schedule times for the p93791 3D-SIC with and without floor planning. The schedule time was shorter when floor planning was applied.

IV. CONCLUSION

In conclusion, this paper proposed a co-optimization technique that integrates thermal-aware test scheduling and floor planning for 3D-SICs, addressing resource and power constraints to minimize total test schedule time. The method effectively reduces both test time and power consumption for the entire 3D-SIC while adhering to temperature and resource limitations. This integrated approach of test scheduling with floor planning surpasses the test scheduling approach without floor planning, proving that the proposed method results in enhanced and optimized testing for 3D-SICs. To address the limitations of the greedy algorithm in thermal-aware floor planning, future research could explore alternative algorithms with better global optimization capabilities, such as Genetic Algorithms, Simulated

Annealing, or Particle Swarm Optimization. Additionally, integrating more established floor planning techniques with the test scheduling method could enhance heat dissipation and overall system performance.

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TABLE I. COMPARISON OF TEST TIME FOR P93791 3D-SIC

TAM	Test schedule time without floor planning	Test schedule time with floor planning
16	1804033	1803812
24	1209389	1208901
32	1023984	1022134
40	778902	775981
48	619001	617410
56	520930	519781